

Dana Martinez Gonzalez

(559) 630-2682 | danapaolawork17@gmail.com | Portfolio: danapaolawork17-creator.github.io

PROFILE

Mechanical engineering student in a BS/MS program with an aerospace minor and three NASA internships across rotorcraft aeromechanics, test engineering, and X-59 aerostructures. Hands-on with thermal and vibration design, structural analysis, Python automation, and CAD in PTC Creo and SolidWorks.

EDUCATION

California State University, Fresno – Lyles College of Engineering

Bachelor of Science in Mechanical Engineering | Minor in Aerospace Engineering

Master of Science in Mechanical Engineering (Finish in Five Program)

Expected Graduation: Spring 2027 (BS) | Spring 2028 (MS) | GPA: 3.52

WORK EXPERIENCE & RESEARCH

NASA Pathways Student Trainee, NASA Armstrong Flight Research Center, Edwards, CA *August 2026 – December 2026*

- Analyze X-59 QueSST flight loads in the Aerostructures Branch alongside Lockheed Martin engineers, supporting a quiet supersonic aircraft built to help lift the ban on supersonic flight over land.
- Ran stress analysis on a modified X-59 engine mount and am preparing a Flight Loads Lab test to validate the modification before the next phase of flight testing.
- Built a CAD assembly for a TG-14 antenna supporting X-59 Phase 2 using SolidWorks and PTC Creo.
- Monitor real-time loads data in the control room during flight tests and write Python scripts to process flight test and structural data.

Test Engineering Intern, NASA Armstrong Flight Research Center, Edwards, CA

June 2026 – August 2026

- Designed the thermal stack for a quantum sensor's 532 nm laser (TEC, copper heat spreader, aluminum optics plate), selecting a 75 W TEC to handle the laser's 12 W max heat load.
- Designed a laser chassis in PTC Creo from measured laser geometry, validated fit with 3D-printed prototypes, and prepared the model for machining (fillets, screw hole tolerances) before ordering through Xometry.
- Selected LORD HT0-2 vibration isolators with a 22 Hz natural frequency, placing resonance below the 150 to 1,000 Hz energy band of a 17.1 Grms flight vibration profile.
- Co-wrote a 73-page random vibration test procedure covering 3 axes, isolated and hard-mounted configurations, and a six-level ramp to 17.1 Grms per AFOP-7900.3-004.

Rotorcraft Aeromechanics Intern, NASA Ames Research Center, Moffett Field, CA

June 2025 – August 2025

- Developed a Python based rotor power calculator with a custom GUI using interpolated CAMRAD II power coefficient tables to evaluate rotorcraft performance.
- Automated airfoil slicing from 3D rotor blade geometry in Rhino using rhinoscriptsyntax to support aerodynamic analysis workflows.
- Supported research on next generation vertical flight vehicles by applying CFD, aeroelastic analysis, and wind tunnel testing to assess performance, noise, vibration, and ride quality.

Engineering Intern, Duncan Enterprises (iLoveToCreate), Fresno, CA

December 2023 – May 2025

- Created packaging specifications and engineering drawings for new product lines using SolidWorks and ArtiosCAD.
- Conducted time studies and data analysis to support production improvements and efficiency reporting.
- Wrote standardized packaging instructions and collaborated with Operations and Quality leadership.

PROJECTS & LEADERSHIP

Chassis and Propulsion Team Member, MATE ROV Competition, Fresno State

November 2025 – Present

- Designed and evaluated a six thruster propulsion layout to enable full 6 DOF control and station keeping in flowing water.
- Contributed to chassis layout, thruster placement, and shroud considerations to improve stability, maneuverability, and safety; developed tether routing and strain relief concepts to protect onboard electronics.

President, Society of Women Engineers (SWE), California State University, Fresno

June 2026 – May 2027

- Lead a team of seven chapter officers, overseeing programming, professional development events, and K-14 outreach initiatives.
- Direct the chapter's WE26 national conference sponsorship campaign, managing corporate outreach, a tiered sponsorship packet, and officer task assignments to fund student travel to Boston.

TECHNICAL SKILLS

Engineering & Analysis: Systems Engineering | Structural & Stress Analysis | Thermal Management | Vibration Analysis | CFD | FEA | Numerical Methods | Engineering Testing (MTS) | Environmental Testing (Thermotron chambers) | Controls

Software & Tools: PTC Creo | SolidWorks | Siemens NX | ANSYS | MATLAB | Simulink | AutoCAD | Revit | SketchUp | ArtiosCAD

Hardware & Systems: RF Systems | PCB Design | Arduino | Soldering | TEC Controllers | Data Acquisition | Instrumentation

Programming: Python | MATLAB | C++

Manufacturing: CNC Machining | Manual Machining | 3D Printing